Day 6 — *Anurūpyeṇa*: Extending to ×12, ×13, …, ×19

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will apply Tirthaji's *Anurūpyeṇa* sub-sūtra ("proportionally") to multiply any 2-digit number by 12, 13, …, 19, with at least 80% accuracy on a mixed worksheet.

Materials

- Whiteboard.
- `activities/activity-02-anurupyena.md` worksheet.
- Stopwatch.

Vocabulary introduced today

- *Anurūpyeṇa* (IAST: anurūpyeṇa; lit. "proportionally") — a sub-sūtra in Tirthaji's corpus. We use it as the name for the proportional scaling that extends the ×11 pattern to ×12, ×13, …, ×19.

Lesson flow

Warm-up (5 min)

- Recall: state the ×11 trick in one sentence.
- *"Outer digits unchanged; middle is the digit sum."*

Core (18 min) — The scaling pattern

1. **Question.** *"What does ×12 look like? Try 23 × 12 by long multiplication first."*

   ```
   23 × 12 = 23 × 10 + 23 × 2 = 230 + 46 = 276.
   ```

2. **Derive.** *"Now distribute (10a + b) × 12 = (10a + b)(10 + 2):*

   ```
   = 100a + 10b + 20a + 2b
   = 100a + 10(b + 2a) + 2b
   ```

   Hmm — that's awkward. *2b* in the units, *(b + 2a)* in the tens. Let's organise it more cleanly.

3. **The Tirthaji pattern.** For N in {11, 12, …, 19}, write $N = 10 + k$ where $k \in \{1, 2, …, 9\}$. Then:

   ```
   (10a + b) × (10 + k)
       = 100a + 10b·k + 10ka + bk
                Wait — let's redo carefully:
       = (10a + b) × 10 + (10a + b) × k
   ```

```
 = 100a + 10b + 10ak + bk
 = 100a + 10(b + ak) + bk
```

Hmm — *bk* could be 2 digits. The pattern is: **scale each digit by k as you slide along.**

Easier form for students: For ×*N* = ×(10 + *k*):

- Start with the leftmost digit.
- Multiply it by *k*, add the next digit, write the units of that sum (carry as needed).
- Continue right.
- The rightmost step: multiply the last digit by *k*.

This is the **Tirthaji form of the *Anurūpyeṇa* extension**.

4. **Worked example: 23 × 13. (k = 3.)**

```
Set up: 0  2  3  (pad with 0 on left to track the carry-out)
Right to left:
  - Units: last digit × k = 3 × 3 = 9. Write 9.
  - Tens: middle digit × k + next-right digit = 2 × 3 + 3 = 9.
Write 9.
  - Hundreds: leftmost digit × k + next-right = 0 × 3 + 2 = 2.
Wait — we want the LEFTMOST as just itself with the carry...
```

(Pause — this is getting confusing. Let's use a different presentation.)

5. **Cleaner presentation for ×12 to ×19.**

For *N* × 12 of a 2-digit number *ab*: simply use long multiplication mentally, but in the order:

- units of answer = *b* × 2 (carry if ≥ 10)
- tens of answer = *a* × 2 + *b* (plus carry; carry if ≥ 10)
- hundreds of answer = *a* (plus carry)

For *N* × 13: replace ×2 with ×3 everywhere.

General: **For ×(10 + k), multiply each digit by k and add the digit to its right (the "running scan").**

6. **Examples** (boxed):

```
23 × 12:
  units: 3·2 = 6
  tens: 2·2 + 3 = 7
  hundreds: 2 (no carry)
  → 276 ✓

23 × 13:
  units: 3·3 = 9
  tens: 2·3 + 3 = 9
  hundreds: 2
```

```
  → 299 ✓

45 × 13:
  units: 5·3 = 15 → write 5, carry 1
  tens: 4·3 + 5 + 1 = 18 → write 8, carry 1
  hundreds: 4 + 1 = 5
  → 585 ✓
```

Check 45 × 13: 45 × 10 + 45 × 3 = 450 + 135 = **585**. ✓

7. **The name.** *"Tirthaji (1965) calls this proportional scaling **Anurūpyeṇa** — 'proportionally.' The ×11 case is the special case k = 1, where the scaling is trivial."*

Activity (15 min) — Mixed drill

See `activities/activity-02-anurupyena.md`.

- 16 problems across ×12, ×13, ×14, ×17, ×19.
- Paired self-check.

Wrap-up (2 min)

- *"How does ×11 fit into the ×12–×19 family?"* (It's the k=1 special case.)
- Preview Day 7: *"What about ×23? ×47? When does the trick stop helping?"*

Student-facing content

> **The ×12 to ×19 family — *Anurūpyeṇa***
>
> For any $N = 10 + k$ (where k is 1 to 9), multiply a 2-digit number $ab \times N$ by: 1. Units = $b \times k$ (carry if ≥ 10). 2. Tens = $a \times k + b$ (plus any carry; carry if ≥ 10). 3. Hundreds = a (plus any carry).
>
> When $k = 1$ (i.e. ×11), this is the original "split and add" trick.
>
> Tirthaji (1965) names this scaling pattern ***Anurūpyeṇa*** — "proportionally."

Homework

- 12 problems mixing ×12, ×13, ×14: see worksheet attached. Self-check with long multiplication on any 4.

Differentiation

- **Tier up:** Try 3-digit × 13 problems. Use the right-to-left running-scan from Day 4 with the *k*-scaling from today.
- **Tier down:** Focus only on ×12 today. Build fluency before adding higher *k*.

Teacher notes

- *Anurūpyeṇa* is the pattern-naming step. Some students will object: *"This is just long multiplication done right-to-left."* They are correct. Honour the observation; reframe: *"The win is doing it MENTALLY in one pass, with a clear rhythm."*

- The procedural complexity grows quickly with k. ×17 and ×19 are harder than ×11 by a real margin. Calibrate the problem set accordingly.
- This day is genuinely harder than Days 1–5. If the class is shaky, slow down — skip the harder k values and revisit on Day 9.

Citation(s) used in this lesson

- Tirthaji (1965), sub-sūtra *Anurūpyeṇa*.
- *Microsoft Word - sutras1.pdf* — for the sub-sūtra list.